

FA7 Part 1

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```
# Stopwatch data
wait_time <- c(6, 15, 3, 5, 11, 9, 6, 4, 23, 19, 34, 8, 3, 18, 16, 7, 4, 4, 4, 6, 8, 41, 38, 19, 3, 14, 4)

# Mean and lambda (rate)
mean_wait <- mean(wait_time)
lambda <- 1/mean_wait

# Print
print(paste0("Total students:", length(wait_time)))

## [1] "Total students:30"

print(paste0("Average gap between students:", round(mean_wait, 2), "seconds"))

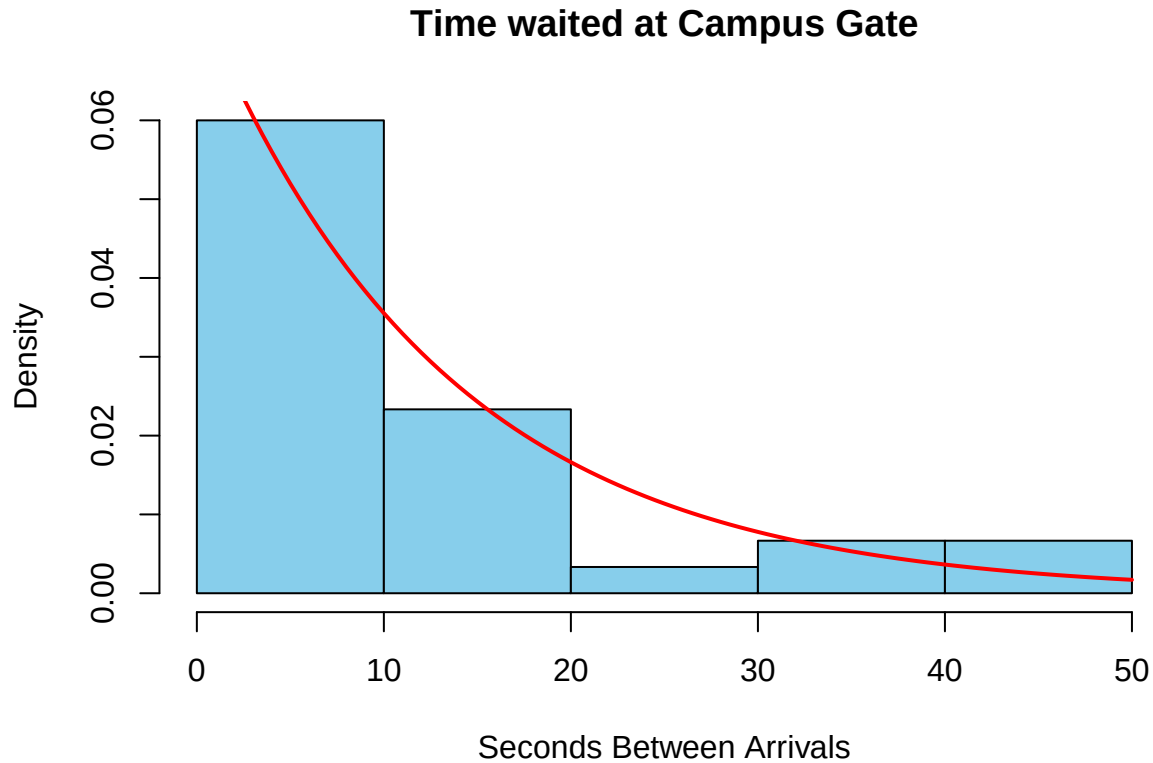
## [1] "Average gap between students:13.17seconds"

print(paste0("Lambda (arrival rate):", round(lambda, 5)))

## [1] "Lambda (arrival rate):0.07595"

#Plot histogram of wait times
hist(wait_time, freq=FALSE, col = "skyblue",
      main = "Time waited at Campus Gate",
      xlab = "Seconds Between Arrivals")

curve(dexp(x, rate = lambda), add = TRUE, col = "red", lwd = 2)
```



Verification of the Exponential Distribution:

The exponential distribution is applicable to this scenario because student arrivals at the campus gate satisfy the two core assumptions:

1. **Randomness:** Students arrive individually based on their own personal class schedules and habits, making the exact timing of each arrival random.
2. **Independence:** One student passing through the gate does not cause or prevent another student from arriving; each arrival event happens independently of the others.

Key Parameters and Formulas :

- **Mean (Expected Value):** $E(X) = \frac{1}{\lambda} = 13.17$ seconds
- **Lambda (Rate Parameter):** $\lambda = 0.07595$ arrivals per second
- **Probability Density Function (PDF):**

$$f(x) = \lambda e^{-\lambda x} = 0.07595 e^{-0.07595x}, \quad x \geq 0$$

- **Cumulative Distribution Function (CDF):**

$$P(X \leq x) = 1 - e^{-\lambda x} = 1 - e^{-0.07595x}$$

Interpretation of Results :

- **Likelihood Calculation:** Using our cumulative distribution function, we can determine the probability that a new student arrives at the campus gate within **10 seconds** of the previous arrival:

$$P(X \leq 10) = 1 - e^{-0.07595 \times 10} = 1 - e^{-0.7595} \approx 0.5321$$

This means there is a **53.21% chance** that the gap between any two arriving students is 10 seconds or less.

- **Practical Implications:** The expected average waiting time between student arrivals is **13.17 seconds**. Because the data follows an exponential distribution, short gaps are highly frequent (as shown by the tall histogram bars between 0 and 10 seconds), while long gaps over 30 seconds are rare outliers. For campus operations, this predictable flow suggests that security or check-in staff at the gate will experience continuous, rapid individual arrivals rather than sudden large batches of students all at once.

Report Findings and Summary :

- **Chosen Scenario:** This study analyzes the arrival patterns of students entering our university campus through **Gate 4**. The objective was to determine whether the time intervals between individual arrivals follow an exponential distribution model.
- **Data Collection Process:** Data was collected on an afternoon by sitting directly near Gate 4. A smartphone stopwatch application was utilized to log precise timestamps, hitting the “Lap” feature every time a student passed through the gate. To capture a natural, unforced flow of arrivals rather than a highly regimented rush hour, observations began at **2:30 PM**, exactly 30 minutes before the major **3:00 PM** classroom transitional period. In total, a sample size of **30 consecutive data points** was successfully recorded.
- **Computed Parameters:** The calculated statistical metrics for this sample show an **Average Waiting Time ($E(X)$) of 13.17 seconds** between student arrivals. This yields an estimated arrival rate (**Lambda, λ**) of **0.07595 students per second**.
- **Interpretation and Real-World Implications:** The empirical histogram aligns very smoothly with the theoretical exponential probability density curve. This strongly indicates that student arrivals at Gate 4 occur randomly and independently during this pre-rush period. The model shows a **53.21% probability** that a new student will arrive within 10 seconds of the previous one. Practically, this tells campus security and administrative staff that gate traffic during this period is a steady, continuous stream of individual arrivals rather than sudden clustered groups, allowing for smooth, manageable entry processing.